

Li-Fi: THE FUTURE OF Modern Warfare Communication

“In modern warfare, secure and undetectable communication is no longer a luxury but a necessity. Li-Fi makes it virtually undetectable, highly secure, and resistant to jamming.” — SLF Telecommunication



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As militaries test and deploy the technology, an important question emerges: Could the next decisive edge in warfare come not from weapons, but from how forces communicate?

Radio frequency (RF) networks are increasingly exposed to interference, blocking, and interception. Li-Fi, or light fidelity, helps overcome many of these risks for the Army, Navy, and Air Force. Because light cannot penetrate physical barriers such as walls, Li-Fi signals remain confined to the operational area. Wi-Fi networks, in contrast, pass through such barriers easily, making remote interception easier for adversaries.

Li-Fi uses the visible light spectrum, 400-800THz. A number of experiments and trials have been conducted, and deployment has begun on the ground.

The technology uses high-speed modulation of LED or laser sources, switching

at nanosecond intervals to create binary data streams that can carry vast amounts of information. These high-speed links remain virtually invisible within the electromagnetic spectrum of the battlefield and provide a secure channel for mission-critical data, from submarine hulls to the silent coordination of autonomous drone swarms.

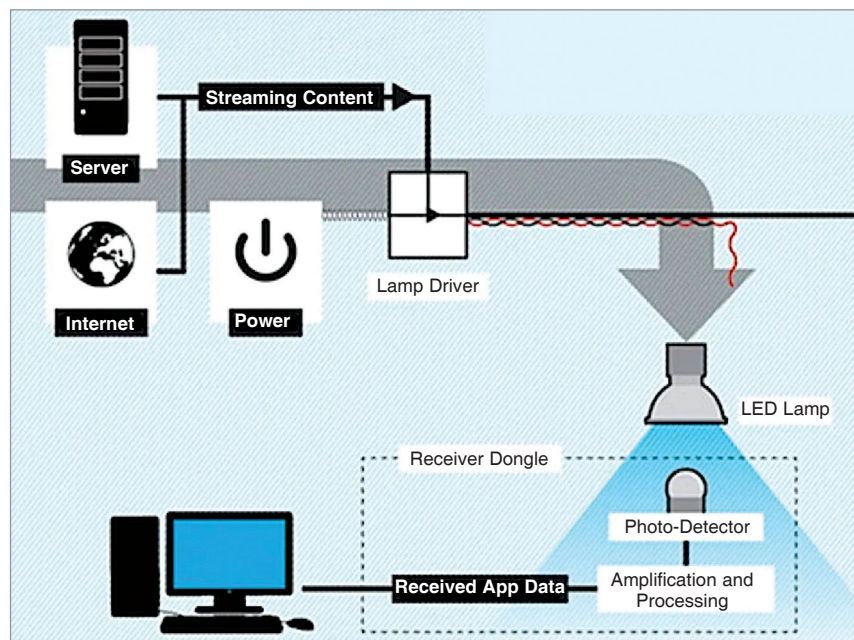
Experiments in the early 2010s demonstrated the suitability of Li-Fi as a channel for secure, short-range, high-capacity communication in restricted environments such as bunkers, depots, and command centres where RF emissions posed risks (Table 1). In contested environments where RF networks are vulnerable, Li-Fi provides covert and interference-resistant communication.

By the mid-2020s, defence agencies worldwide began adapting Li-Fi technology for dynamic platforms such as drone swarms, underwater teams, and armoured vehicles operating in diverse domains.

Building indigenous Li-Fi systems for defence

Communication signals in border areas, no-man's-land, mountainous regions, and naval bases are prone to interference, blocking, or eavesdropping because they rely on radio and satellite links that remain vulnerable. The danger increases further as some countries deploy advanced electronic warfare capabilities.

The National Research and Education Network of India (ERNET), in partnership with IIT Madras, conducted feasibility studies that confirmed the need for Li-Fi to reduce these risks. Encouraged by these results, India has initiated



How Li-Fi works (Source: pureLiFi)